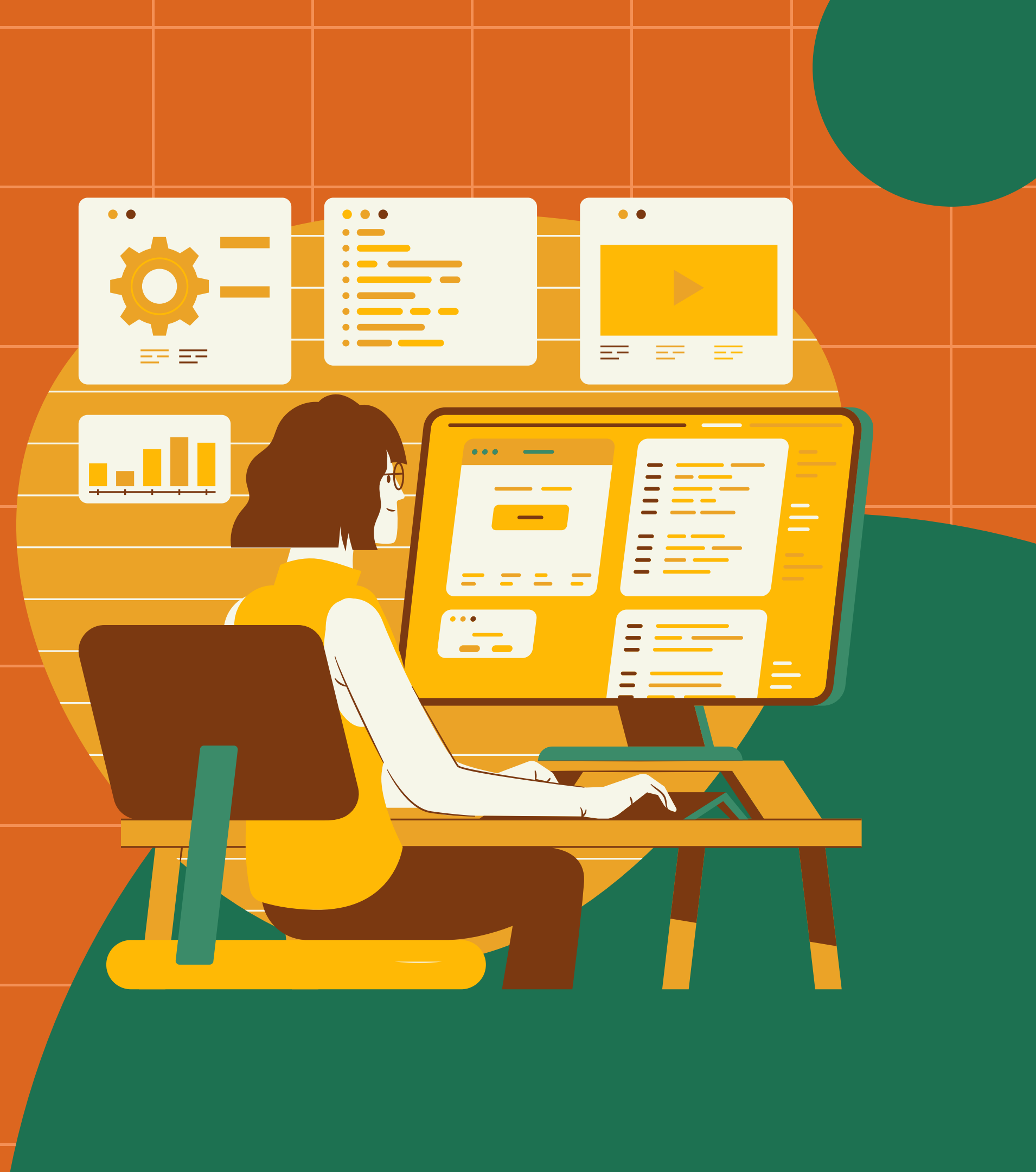


AUTOMATED CUSTOMER REVIEWS

SENTIMENT CLASSIFICATION, PRODUCT CLUSTERING,
AND GENERATIVE SUMMARIZATION

PRESENTED BY: LAMA ALWUTHAYNANI



DATASET COLLECTION AND PREPROCESSING



Datasets Used:

- Two Kaggle datasets with identical structure.
- Appended together using `pd.concat()` to create a unified dataset.

Preprocessing Steps:

- Removed duplicates and null values
- Merged title + text into one field
- Mapped ratings:
 - ★ 1-2 → Negative
 - ★ 3 → Neutral
 - ★ 4-5 → Positive

Class Distribution:

Balanced by downsampling positive reviews to improve model training

Review Classification

Classify customer reviews into positive, negative, or neutral categories

Model Architecture

Model: roberta-base (Transformer from Hugging Face)

Tokenizer: RobertaTokenizerFast

Custom loss with class weights to handle imbalance:

TrainingArguments

- 3 Epochs, Batch Size: 8, LR: $2e-5$
- Evaluation after each epoch
- Used Hugging Face Trainer with custom `compute_loss()` and `compute_metrics()`
- Split data: 80% training, 20% testing



Evaluation Results

Model achieved an accuracy of 89.64% on the test dataset.

Class Positive:

Precision: 95.08%
Recall: 95.75%
F1-Score: 95.41%

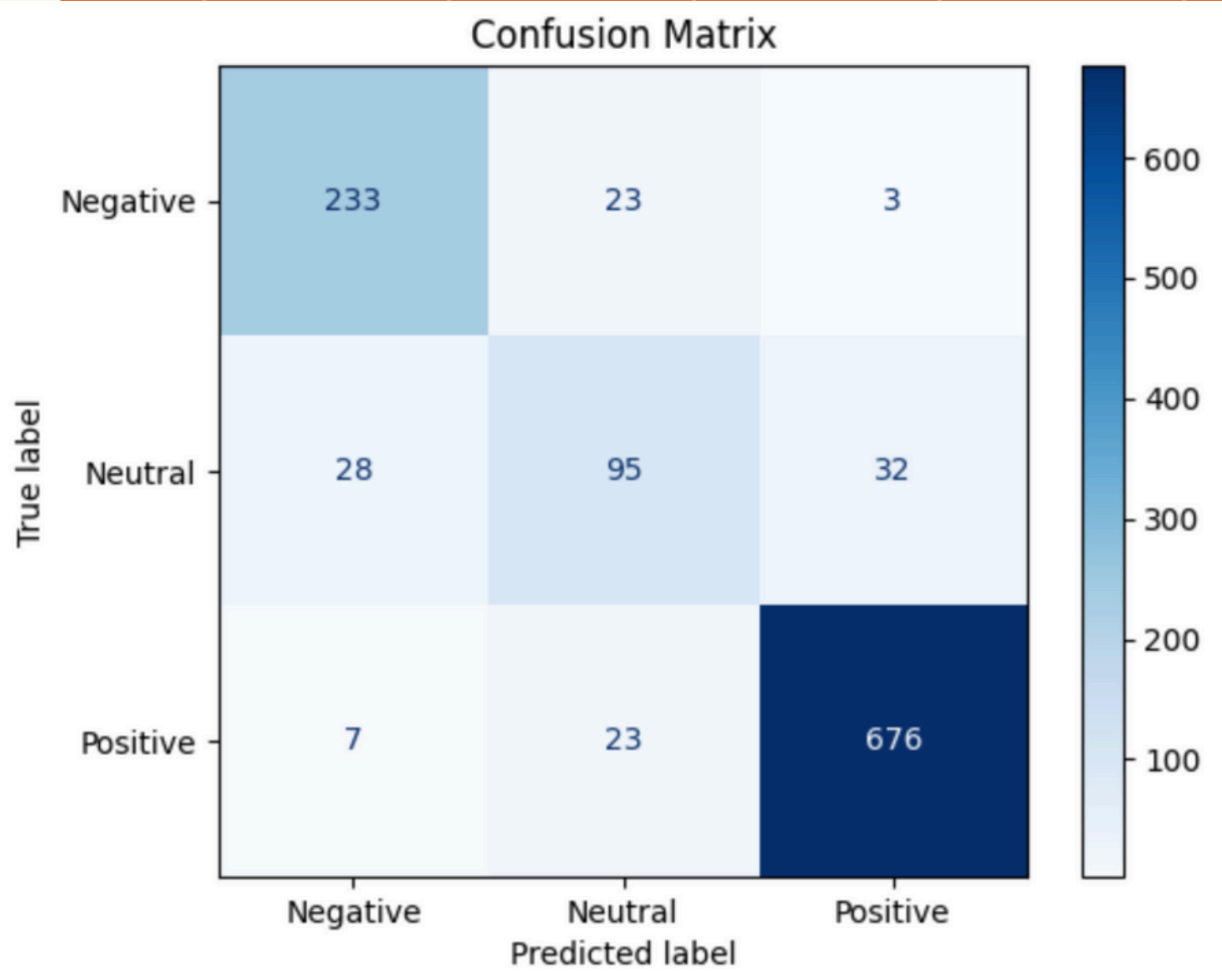
Class Neutral:

Precision: 67.38%
Recall: 61.29%
F1-Score: 64.19%

Class Negative:

Precision: 86.94%
Recall: 89.96%
F1-Score: 88.43%

Confusion Matrix



PRODUCT CATEGORY CLUSTERING

Group products into broad, meaningful clusters

Extracted main category from:

- categories
- primaryCategories

Standardized and mapped categories to 5 meta-groups:

- Ebook readers
- Electronics
- Accessories
- Batteries
- Home & Office
- etc

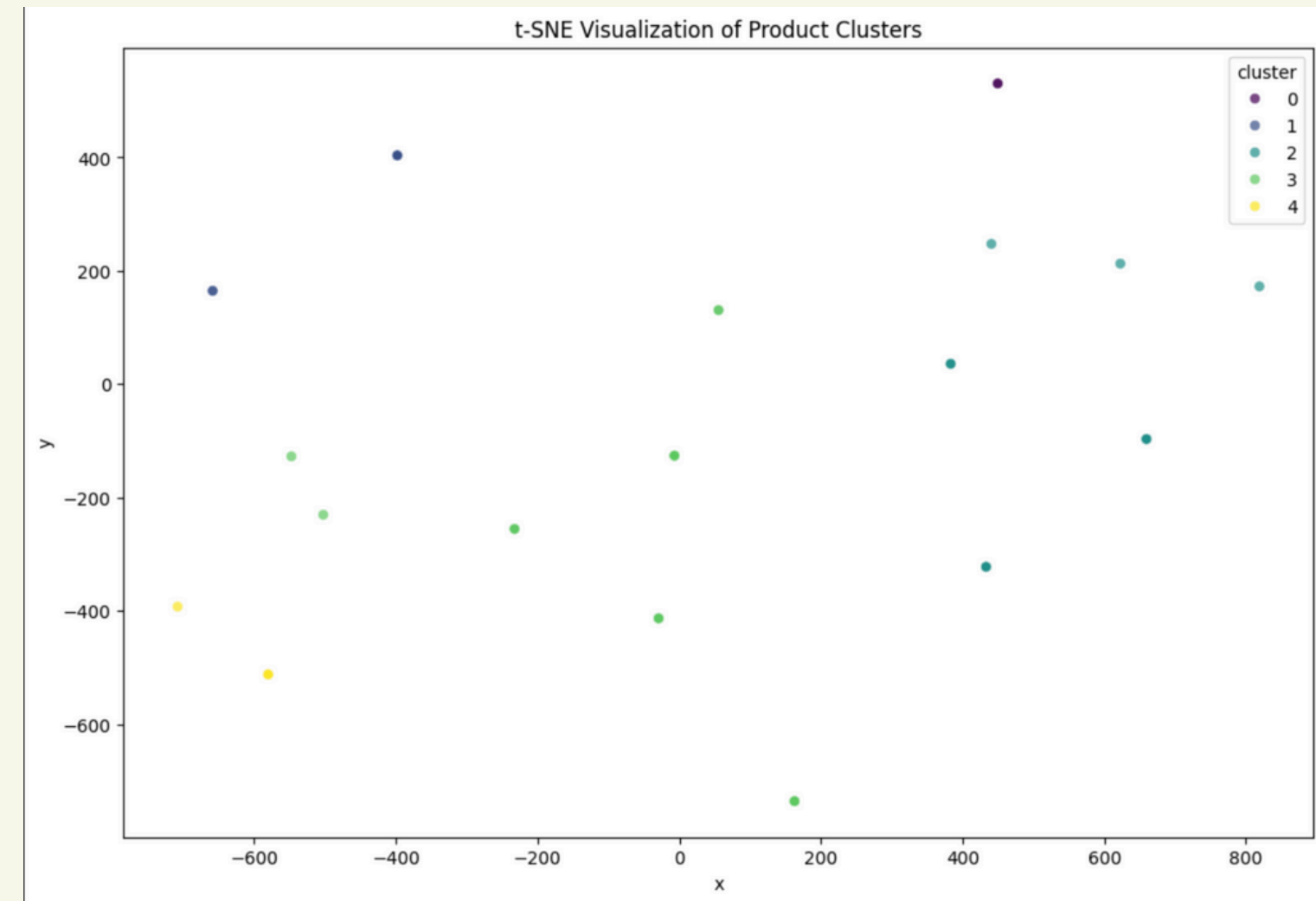


EMBEDDING & CLUSTERING

- Used Sentence-BERT to convert category text into embeddings.
- Applied KMeans with k=5 clusters.

Evaluation Metrics:

Silhouette Score: 88%



REVIEW SUMMARIZATION WITH GENAI

Generate short blog-style articles that recommend top products per category based on customer reviews

Output per Article:

- Top 3 products with key highlights
- Complaints for each top product
- 1 Worst product and why to avoid it

Pretrained model: facebook/bart-large-cnn

PROCESS OVERVIEW

Data Preparation

- Merge reviews with product clusters
- Filter by rating & sentiment (Positive/Negative)

Analysis:

- Pick top-rated (≥ 5 reviews) and worst-rated
- Extract up to 10 reviews per sentiment for each product

Article Generation

- Use BART to summarize reviews
- Output: Markdown article per cluster

DEPLOYMENT

Platform: Streamlit Cloud

1. Review Classifier

Input a review → Get sentiment (Positive / Neutral / Negative)

2. Review Dashboard

- Search any product
- View sentiment stats & top reviews

Scan to Explore:



SCAN ME

Streamlit - A faster way to build ML apps

Review Intelligence Dashboard

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Share ☆ ✎ ↺ ⋮

app

Review Classification

Review Intelligence Das...

Review Intelligence Dashboard

🔍 Enter Product Name:



THANK YOU